14 Lecture 14 (February 14, 2025, est.): Relative Tannaka duality

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14.1 Changing the target of a fiber functor

The next goal is to study symmetric tensor categories by realizing them as categories of representations of affine group schemes in a suitable symmetric tensor category. Ordinary Tannaka duality uses Vect_k as the target; the relative version allows a different target $\mathcal{D}$.

Fix a perfect field k. In this lecture, a symmetric tensor category is essentially small, k-linear, abelian and rigid, with finite-dimensional morphism spaces, objects of finite length, and $\mathrm{End}(\mathbb{1}) = k$. A tensor functor is exact, k-linear and strong symmetric monoidal. Such a functor is faithful. Write Symm_k for the category of these categories and isomorphism classes of tensor functors. Its full subcategory of categories of moderate growth consists of those $\mathcal{C}$ for which, for each $X \in \mathcal{C}$, there is a real number $a_X \geq 1$ satisfying

$$\mathrm{length}(X^{\otimes n}) \leq a_X^n \qquad (n \geq 1).$$

Fix $\mathcal{D} \in \mathrm{Symm}_k$. An object of $\mathrm{Symm}(\mathcal{D})$ is a pair $(\mathcal{C}, F)$ with $F : \mathcal{C} \to \mathcal{D}$ a tensor functor. A morphism from $(\mathcal{C}_1, F_1)$ to $(\mathcal{C}_2, F_2)$ is represented by a pair (T, α), where $T : \mathcal{C}_1 \to \mathcal{C}_2$ is a tensor functor and $\alpha : F_1 \xrightarrow{\sim} F_2 T$ is a monoidal natural isomorphism. We identify two such pairs when there is a compatible monoidal natural isomorphism between their functors T. Because F_2 is faithful, a compatible isomorphism between two fixed pairs is unique if it exists. This does not say that the morphisms of $\mathrm{Symm}(\mathcal{D})$ themselves are unique.

14.2 The internal group of tensor automorphisms

An affine group scheme in $\mathcal{D}$ is described by a commutative Hopf algebra in $\mathrm{Ind}(\mathcal{D})$. The ind-completion is necessary: a coordinate algebra need not be an object of finite length. If $F : \mathcal{C} \to \mathcal{D}$ is a tensor functor, the coefficient construction gives a commutative Hopf algebra

$$A_F = \int^{X \in \mathcal{C}} F(X)^* \otimes F(X) \quad \text{in } \mathrm{Ind}(\mathcal{D}).$$

Here the integral denotes a coend. Concretely, its structure maps $i_X : F(X)^* \otimes F(X) \to A_F$ satisfy, for every $f : X \to Y$,

$$i_X\big(F(f)^* \otimes \mathrm{id}_{F(X)}\big) = i_Y\big(\mathrm{id}_{F(Y)^*} \otimes F(f)\big) \quad \text{on } F(Y)^* \otimes F(X).$$

It is universal for families with these relations. Tensor products in $\mathcal{C}$ define multiplication, the unit object defines the unit, evaluation and coevaluation define the coalgebra structure, and duality defines the antipode. Symmetry makes the multiplication commutative. This is the internal analogue of the matrix-coefficient Hopf algebra.

14.1 Definition. The affine group scheme represented by A_F is denoted $\underline{\mathrm{Aut}}^{\otimes}(F)$. The *fundamental group* of $\mathcal{D}$ is

$$\pi(\mathcal{D}) = \underline{\mathrm{Aut}}^{\otimes}(\mathrm{Id}_{\mathcal{D}}).$$

The universal coefficient coactions make every object of $\mathcal{D}$, and then every ind-object, a representation of $\pi(\mathcal{D})$. Naturality means that every morphism in $\mathcal{D}$ respects these canonical actions. Applying the canonical action to the objects $F(X)$ gives a homomorphism

$$\psi_F : \pi(\mathcal{D}) \longrightarrow \underline{\mathrm{Aut}}^{\otimes}(F).$$

14.2 Example. We have $\pi(\mathrm{Vect}_k) = 1$. Indeed, naturality with respect to maps $k \to V$, together with the tensor-unit condition, forces a tensor automorphism of the identity to be the identity on every vector space. If $\operatorname{char} k \neq 2$, then $\pi(\mathrm{SVec}_k) = \mu_2 \cong \mathbb{Z}/2$, regarded as an internal group concentrated in even degree. Its nontrivial element acts as $+1$ on the even part and -1 on the odd part. This is the parity automorphism, not the functor that exchanges even and odd degrees.

14.3 Compatible groups and representations

Not every homomorphism from $\pi(\mathcal{D})$ to an internal group is compatible with the structure already present in $\mathcal{D}$.

14.3 Definition. A *$\mathcal{D}$-group* is a pair (G, ϕ), where G is an affine group scheme in $\mathcal{D}$ and $\phi : \pi(\mathcal{D}) \to G$ is a homomorphism such that the canonical action of $\pi(\mathcal{D})$ on $\mathcal{O}(G)$ equals the action induced by conjugation in G along ϕ. A morphism $(G, \phi) \to (H, \psi)$ is a homomorphism $u : G \to H$ with $u\phi = \psi$.

For clarity, put $A = \mathcal{O}(G)$ and $B = \mathcal{O}(\pi(\mathcal{D}))$. Using left coactions, the compatibility condition says that the canonical map $A \to B \otimes A$ is the composite

$$A \xrightarrow{\Delta^{(2)}} A \otimes A \otimes A \xrightarrow{\phi^* \otimes \mathrm{id} \otimes S_B \phi^*} B \otimes A \otimes B \xrightarrow{(m_B \otimes \mathrm{id})(\mathrm{id} \otimes c_{A,B})} B \otimes A.$$

The last arrow uses the symmetry to bring the two B factors together. The pair $(\underline{\mathrm{Aut}}^{\otimes}(F), \psi_F)$ is a $\mathcal{D}$-group.

14.4 Definition. For a $\mathcal{D}$-group (G, ϕ), let $\mathrm{Rep}_{\mathcal{D}}(G, \phi)$ be the full subcategory of representations of G on objects of $\mathcal{D}$ whose restriction along ϕ is the canonical $\pi(\mathcal{D})$-action. It has a forgetful tensor functor $\omega : \mathrm{Rep}_{\mathcal{D}}(G, \phi) \to \mathcal{D}$.

14.5 Theorem (Relative Tannaka duality)**.** *The assignments*

$$(\mathcal{C}, F) \longmapsto (\underline{\mathrm{Aut}}^{\otimes}(F), \psi_F), \qquad (G, \phi) \longmapsto (\mathrm{Rep}_{\mathcal{D}}(G, \phi), \omega)$$

give mutually inverse contravariant *equivalences between* $\mathrm{Symm}(\mathcal{D})$ *and the category of $\mathcal{D}$-groups.*

This is the relative reconstruction theorem; see [CEO24, §4, Theorem 4.2.1]. Its full reconstruction proof is not part of this lecture. The variance can be checked directly: a pair (T, α) as above sends an automorphism ξ of F_2 to

$$\alpha^{-1}(\xi T)\alpha \in \underline{\mathrm{Aut}}^{\otimes}(F_1).$$

Thus a functor between representation categories produces a group homomorphism in the opposite direction. For $\mathcal{D} = \mathrm{Vect}_k$, the fundamental group is trivial and we recover ordinary neutral Tannaka duality.

14.6 Example. For $\mathcal{D} = \mathrm{SVec}_k$, $\operatorname{char} k \neq 2$, an internal group is an affine supergroup. A compatible map $\mu_2 \to G$ is specified by an even element $z \in G(k)$ with $z^2 = 1$ whose conjugation action is the parity automorphism of $\mathcal{O}(G)$. The category $\mathrm{Rep}(G, z)$ consists of representations on which z acts by parity. If G is purely even, the condition makes z central. For a general supergroup, centrality must not replace the parity-conjugation condition.

14.4 The characteristic-zero classification

Now suppose that k is algebraically closed of characteristic zero. Deligne's theorem states that a symmetric tensor category admits a fiber functor to SVec_k if and only if it has moderate growth. Consequently it is of the form $\mathrm{Rep}(G, z)$ above. The fiber functor, when it exists, is unique up to monoidal natural isomorphism; the isomorphism need not be unique. These statements apply to arbitrary essentially small pretannakian categories; no tensor-generation hypothesis is imposed. For this formulation and uniqueness, see [CEO24, §§1.1, 1.5]. In the ordinary Tannakian case, uniqueness without a finite-generation hypothesis also follows from [Wib25, Corollary 1.2].

The ordinary Tannakian criterion is stronger: there is a fiber functor to Vect_k if and only if, for every object X, some exterior power $\bigwedge^{n_X} X$ vanishes. The bound may depend on X. The next lecture explains why arbitrary Schur functors replace exterior powers in the super version of this criterion.

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